



Read and Change a Loop

Ontario Math C3.2 — read & alter code

Name: _____

Date: _____

★ I can read a loop, change it, and say what happens.

Read the code first

Good coders READ code before they change it — like reading a recipe before you cook.

Here is one of Bit's loops. Read it, see what it does, then change a number and predict the new result.

Read this loop

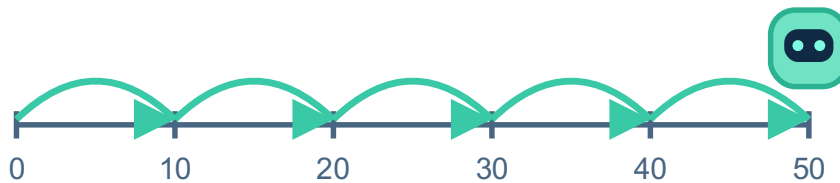
when green flag clicked

repeat 5

move 10 steps

Bit moves 10 steps, 5 times. That is 5 moves and 50 steps in total.

repeat $5 \times \text{move } 10 = 50$ steps



Read it as 5 jumps of 10 → 50 steps.

1 Read it

Look at the loop above. How many times does Bit move? _____ Each move is 10 steps, so how far in TOTAL? _____

2 Change the count, predict

Change repeat 5 to repeat 3. PREDICT: how many moves now, and how far in total? Then check.

3 Change the inside

Now keep repeat 5, but change "move 10" to "move 2". How far does Bit go in total now? How did changing the inside block change the result?

4 Challenge

You want Bit to move 40 steps in total, moving 10 each time. What repeat number do you need? Show how you worked it out.

Big idea

Read a loop to see what it does. Changing the repeat number changes how many times; changing the inside block changes what happens each time.



Read and Change a Loop

Quick facilitation notes & answer key

What this worksheet teaches

Students read a loop to say what it does, then change the repeat number or the inside block and describe how the result changes. There is also a work-backwards challenge. No coding experience needed.

Before you start

No computer needed — the worksheet shows the blocks and what Bit does, so you can teach from the printout. Optional: a Scratch-style app to drag the blocks and watch Bit run — you only press the green flag.

How to lead it (about 30 minutes)

1. Show it (you do). Read the loop: 5 moves of 10 steps make 50 steps in total.
2. Do one together. Exercise 1 — 5 moves, 50 steps.
3. Let them work. Students change the repeat (Ex 2), change the inside block (Ex 3), and work backwards to a repeat number (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — 5 moves; 50 steps total. Exercise 2 — 3 moves; 30 steps.

Exercise 3 — 10 steps (5×2); changing the inside block changed how far each move goes. Exercise 4 (challenge) — repeat 4 ($40 \div 10 = 4$).

If students get stuck — and ways to adjust

Watch for: changing the repeat when the question changes the inside block (or the other way around). Read which part changed.

Make it easier: write "moves $\times$ steps each" and fill in the numbers.

Make it harder: ask the repeat number for 60 steps at 10 each (6).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The child reads the loop, changes one part, and works backwards to a repeat number.